

Name: NGENO KIBET CALEB

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BIT2223: Mobile Computing

ASSIGNMENT

a

- i. It improves internal communication by adding data privacy thus patient records confidentiality unlike traditional telephone systems which may be susceptible to interference/misconfiguration.
- ii. IP telephony uses existing hospital data networks such as integrating into existing IT systems, unlike traditional systems that need separate infrastructure that support voice only communication. Traditional systems depend on cables and circuit switching while IP telephony utilizes TCP/IP decoupling phone extension.
- iii. IP telephony is cheaper because it guarantees free internal calls reducing operational costs unlike traditional systems that would charge per line.

IP telephony allows for mobility unlike traditional systems that limit communication to stations. For instance, nurses/doctors can monitor patients even when they are far with the help of IP telephony. They would need to be at a workstation/booth to communicate using traditional systems.

Since IP telephony can be integrated into IT systems, it allows for automation such as panic/emergency fostering coordination and data privacy. Traditional systems would not guarantee a faster way during emergencies, lacking contextual communication thus using broadcast risking data security and slow reaction.

b.

Wi-Fi calling allows mobile devices to route voice and data traffic over the hospital's wireless LAN (WLAN) infrastructure when cellular signals are down due to high radio frequency signal attenuation resulting from various factors such as a distant cell tower and thick hospital walls.

If not secured, WIFI networks may be abused by hackers leading to losses. Wi-Fi calling requires robust security protocols to prevent unauthorized access.

Wi-Fi calling is susceptible to network congestion. If the hospital's Wi-Fi is not optimized for voice traffic and high bandwidth, it will be affected by latency and quality of service drop.

Wi-Fi networks rely on availability of electricity, therefore there is no service during power outage.

c.

Cellular networks support mobility because they are in portable devices and have wide area network coverage unlike Wi-Fi networks that avail network at a locality usually 10m radius or Wi-Max available in a municipality except the satellite powered. Wi-Fi networks have limited range and may affect link stability during transit. The cost of installing, maintaining and running cellular networks is quite low, because it utilizes the available infrastructure such as mobile technology, unlike deploying Wi-fi networks that is characterized by high cost of installation.

Cellular networks would be suitable for this scenario, ambulance, because it is portable, affordable and supports wide area network coverage.

d.

Handoff is the process in cellular communication where a mobile device's connection is transferred from one base station (cell) to another as the device moves, without interrupting the ongoing/active communication.

It is important because it ensures seamless communication during transit without drop/breakdown. There is no interruption thus quality of communication is maintained allowing the ambulance and doctors/ nurses to keep a line of service active during an emergency.

e.

i) Network segmentation whereby the network is split into different layers supporting different users. Creating a segment for every group of users of the network enables better security.

ii) Implementing a strict password policy to protect network access from unauthorized users. This can be done by enforcing WPA3 security to encrypt network traffic.

iii) Monitoring connected devices by filtering traffic and implementing firewalls to protect the network from attack. Instances can be triaged and anomaly detection done early to ensure security in the network.